

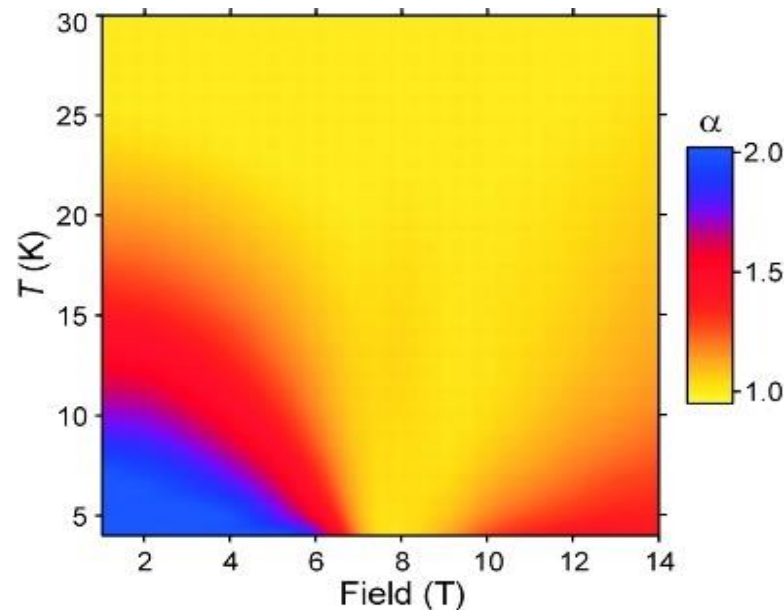
Competing magnetic orders and anisotropic transport in quantum critical $\text{Sr}_3\text{Ru}_2\text{O}_7$

Aditya Putatunda
Qin Guanhua
Wei Ren
David J Singh

University of Missouri-Columbia
APS March Meeting 2020, Denver, Colorado

Quantum critical point and $\text{Sr}_3\text{Ru}_2\text{O}_7$

- Occurs when the T_{PT} is depressed down to absolute zero by tuning
- $\text{Sr}_3\text{Ru}_2\text{O}_7$ shows a first-order metamagnetic transition whose T_{PT} can be depressed towards absolute zero.

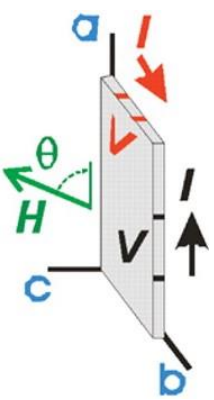


Grigera et al, Science 294, 329 (2001)

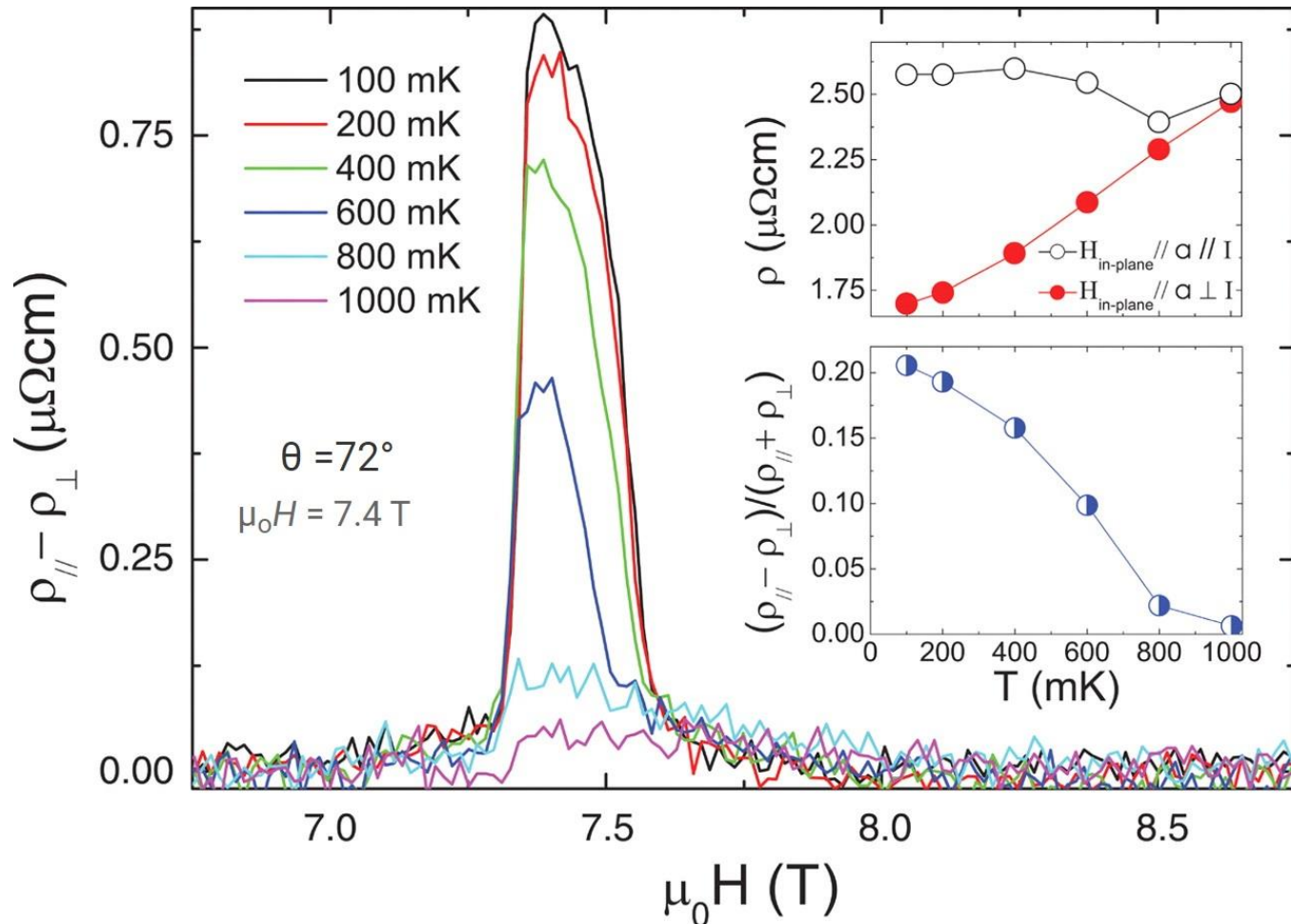
- Fermi liquid theory: $\rho = \rho_{\text{res}} + AT^\alpha$ [with $\alpha=2$ for low T electron-electron scattering]

- Borzi *et al* (2007) reported:
Additional magnetoresistive
anisotropy [Fig.]

Sr₃Ru₂O₇ – Nematicity



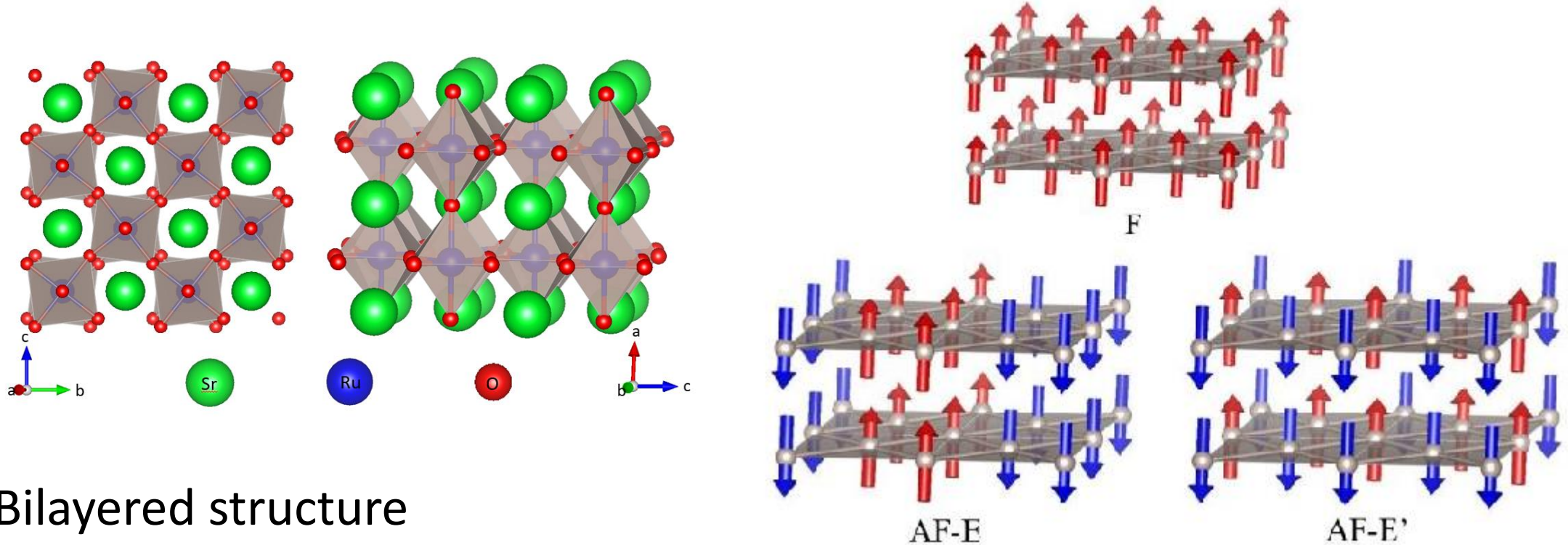
- “Natural Interpretation [...] Sr₃Ru₂O₇ hosts a symmetry broken nematic phase”
-Fradkin, Mackenzie *et al.* (2010)
- Metamagnetism, the QCP and its associated fluctuations are interrelated. Remains to know -> **Nematicity**



Theory & our first-principles approach

- QC material => Strong fluctuations & associated scattering; disruption of the magnetic order.
 - Nature of these fluctuations, determined by =>
Ground state as well as lowest lying metastable states.
 - Fluctuation-dissipation theorem
 - Such metastable states are often seen to be in competition in a QC material.
-
- Searched for low lying AFM orders using DFT calculations; both LDA and PBE functionals.
 - Found a striped magnetic ordered state, closest to the ground state

Results



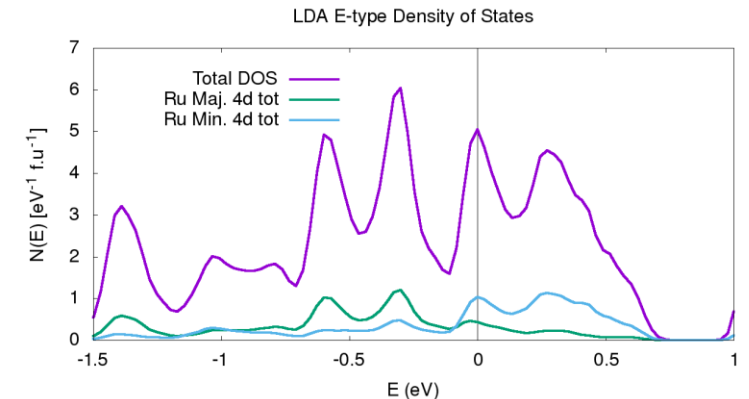
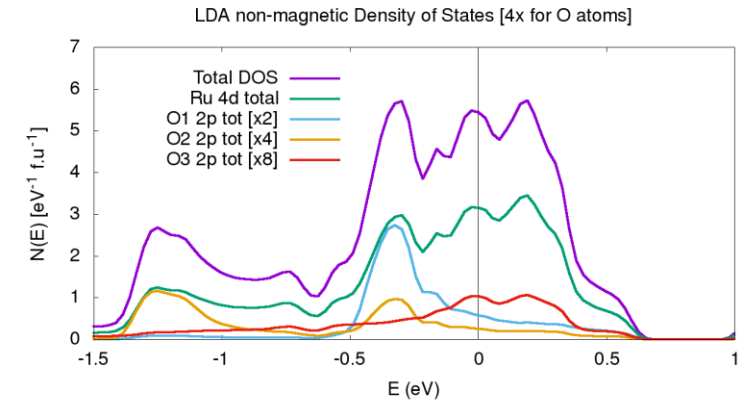
- Bilayered structure
- Experimentally paramagnetic
- Ferromagnetic within DFT [Order destruction]
- Lowest metastable(s): Striped E-type orders [Orthorhombic distortion]

Energetics & electronic structure

- E-type AFM order: lowest metastable
- Itinerant: FM and A-type are different
- Density of states: Broad peak at E_F
- Strong Ru-4d O-2p hybridization

Mesa *et al.* PRB 85 180410(R) [2012]

Order FPLAPW	PBE		LDA	
	$\Delta E(\text{meV})$	Mag mom Ru (μB)	$\Delta E(\text{meV})$	Mag mom Ru (μB)
FM	-147.8	1.28	-29.9	0.73
E	-131.5	1.08	-28.6	0.85
E'	-130.3	1.08	-28.0	0.85
A	-95.6	1.06	-21.2	0.60
C	-43.6	0.72	-13.6	0.38
G	-3.2	0.45	—	—
NM	0	0.000	0	0.000



Transport & its anisotropy

- Solving Boltzmann Transport equations for various orders within CSTA and LDA functional, we found:

Order	Diagonalized σ/τ ($10^{18} (\Omega \text{ m s})^{-1}$)			Anisotropies
E	6.2	65	77	1.17
E'	3.89	67	75	1.13
A	2.54	250	250	1.00
C	1.88	230	233	1.02
FM	2.00	138	144	1.04
	7.24	205	214	1.04
NM	4.08	270	270	1.03

Conclusions

- DFT predicts a FM state against a PM exp. ground state, presumably due to strong fluctuations inherent to a material with QCP
- Energetics search reveal a striped (E-type) AFM ordered phase
- Transport calculations show sizeable anisotropies associated with this phase.
- Inelastic neutron studies investigating such fluctuations in relation to the observed nematicity might be interesting.